Hitender K. Oswal

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OBJECTIVE: to earn a Ph.D. in Computer Science for a career in machine learning in medical research.

EDUCATION

University of Washington, Allen School Sept. 2023 – June 2027

B.S. in Computer Science (GPA 3.9)

- **Relevant Coursework**: Autonomous Robotics, Artificial Intelligence, Computational Biology, Neural Engineering, VR Systems, Advanced Machine Learning
- **Awards**: Dean's List (2023-Present), UW Edward and Theresa McMahon Scholar, Washington State Opportunity Baccalaureate Scholar, Undergraduate Research Conference Travel Award

WORK EXPERIENCE

Amazon Web Services (AWS) Summer 2025 & 2026

Software Development Engineer Intern

- Designed and implemented a scalable microservice for AWS GuardDuty that monitors security threats and vends data to customer accounts globally.

KurtLab, University of Washington May 2024 – Present

Researcher

- Presently developing a medical foundation model for synthetic CT brain generation using NVIDIA's MAISI model
- Previously designed novel brain segmentation models for tumor and stroke lesion segmentation
- **ISLES'24 Challenge Winner**: Applied clinically-informed preprocessing techniques to stroke segmentation models

UW Reality Lab, University of Washington Nov. 2023 – Jun. 2024

Researcher

- Built an interactive Virtual Reality game that aids in the understanding of fundamental computer science concepts and data structures through visualization
- Awarded **Best Developer Tool Award** at DubHacks 2023

RESEARCH PUBLICATIONS

Medical Image Computing

- KurtLab. 2025. "Here Comes the Explanation: A Shapley Perspective on Multi-contrast Medical Image Segmentation." ***The 3rd World Conference on eXplainable AI (XAI 2025)***.
- KurtLab. 2025. "Clinically-Informed Preprocessing Improves Stroke Segmentation in Low-Resource Settings." ***MIRASOL Workshop, MICCAI 2025***.
- KurtLab. 2024. "An Ensemble Approach for Brain Tumor Segmentation and Synthesis." ***MICCAI 2024***.
- KurtLab. 2024. (Challenge Winner) "How We Won the ISLES'24 Challenge by Preprocessing." ***SWITCH Workshop, MICCAI 2024***.

Human-Computer Interaction

- Sushil K. Oswal and Hitender K. Oswal. 2026. "'It Was Frustrating to See These LLMs Glitch': Evaluating Their Workflows and Products for Accessible User Experiences by Novice BLV Users." ***ICCHP 2026***.
- Sushil K. Oswal and Hitender K. Oswal. 2025. "Conversational AI for Accessible Website Design: Integrating LLM Assistants in Website Builders." ***CWUAAT 2025, University of Cambridge/Springer***.
- Sushil K. Oswal and Hitender K. Oswal. 2024. "Examining the Accessibility of Generative AI Website Builder Tools for Blind and Low Vision Users: 21 Best Practices for Designers and Developers." ***IEEE ProComm 2024***.

- Sushil K. Oswal and Hitender K. Oswal. 2023. "Study of a Smartphone App as a Bridge Assistive Technology for COVID-19 Home Test: 19 Essential Guidelines." ***CWUAAT 2023, University of Cambridge/Springer***.
- Sushil K. Oswal and Hitender K. Oswal. 2023. "A Story of Four Documentation Designs for One Medical Device: User Experience Analysis and Recommendations." ***ACM SIGDOC '23***.

Other Publications

- Hitender K. Oswal. 2024. "A Mentoring Model for Promoting Computational Thinking: Understanding Student Leadership and Learning Environment in an Extracurricular Robotics Team." ***para.chi.dub 2024***.
- Hitender K. Oswal, et al. 2024. "Understanding the Leadership Structure and Mentoring Model of an Extracurricular Robotics Team: Key Findings from a Case Study." ***ACM SIGCSE 2024***.

OTHER UW PROJECTS

Extended Reality Association (XRA) Sept. 2025 – Present

President

- Planned and hosted Hack the AM, the first Extended Reality Hackathon hosted at the University of Washington
- Orchestrated spatial computing initiatives and technical workshops for a community of 80 members, facilitating hands-on access to emerging hardware including Meta Quest 3 and Apple Vision Pro.

Composite Backdoors in Reasoning LLMs April 2026 – June 2026

Project for Advanced Machine Learning

- Investigated literature related to backdoors in LLMs and designed a composite backdoor
- Tested the effectiveness of composite backdoors on modern reasoning LLMs by finetuning and evaluting 6 open-source LLMs

Adaptive Thermal Haptics for Virtual Reality Jan. 2025 – May 2025

Built for VR Systems and VR Capstone

- Designed custom wireless, Peltier-driven wrist-mounted hardware to provide thermohaptic feedback
- Integrated real-time temperature feedback over Bluetooth in Unity for VR environments

SKILLS & MEMBERSHIPS

Skills: Python, Java, C#, PyTorch, TensorFlow, ROS, XR Development

Memberships: Association for Computing Machinery (ACM), IEEE, ACM Puget Sound SIGCHI Chapter